

Manual

Entry/Information Functions for NC Programs/ Setting Data AIP-DNC 8.1

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1 Overview Input/Info Functions for NC Programs/Settings

Purpose

The DNC module makes functions available that are used to manage DNC programs, to download programs onto a machine or to upload revised programs onto a machine.

Implementation considerations

You use the function package if you:

- Want to upload a resource onto a machine or download a resource from the machine.
- Want to display a machine's DNC resource.
- Would like to see documents allocated to a DNC resource.
- Would like to block, release or change the status of a DNC resource.

Features

The Acquisition Information Panel (AIP) offers a wide array of functions for DNC operation and for handling setting parameters to BDE/ MDE terminals, industrial PCs or PCs:

- Shows the NC or the setting data records in a list with numerous options to make selection easier
- The ability to select a data record and to transfer it to the machine control system (download)
- Function with which to transfer a selected or a new data record from the machine control system to HYDRA (upload)
- Status functions (e.g. blocking, releasing) with which to maintain and manage NC programs at the terminal
- Visualization of NC records and setting parameters at the terminal in conjunction with the function to display tooling sheets (DNC-AEB)

2 DNC Functions of AIP

2.1 General information / basic view of DNC

The HYDRA-DNC module provides functions to manage DNC programs, to download DNC programs onto a machine or to upload changed programs from a machine.

Provided that the module is active, it can be opened in the “workplace” section using the “DNC” button. The DNC function is started for the machine selected in the basic dialog. However, the DNC dialog cannot be opened if a machine is selected that is not assigned to a DNC family.

The opened dialog (DNC basic dialog) shows the below-mentioned data.

Operation

Operation number and designation of the operation that is logged on and selected in the basic dialog.

Order

Order number of the operation logged on and selected in the basic dialog.

Article

Article/item number of the operation logged on and selected in the basic dialog.

Workplace

Workplace/machine number and short description.

Selection

By default, selection is made “by DNC family”. The configured filter fields displayed..

But it is also possible to define another type of selection that is specific to the machine within the hytnrcfg.ini file (the entries to be made are written in brackets).

Selection by order (KEYTYP=A)

The “search term” fields, etc. are hidden. The filter field is activated instead within the table. DNC resources are already loaded and displayed while the dialog is being opened. The information line below the grid is hidden.

Selection by program (KEYTYP=P)

The table is empty at first. The “search term” field refers to the “DNC element”. Filtering refers to the DNC element.

DNC family

The DNC family which the currently selected machine is assigned to is displayed here. Please note: A machine should only be assigned to one DNC family. Any of the assigned DNC families will be displayed here if the machine is assigned to more than one DNC family.

“Released” status only

If this checkbox is checked only DNC resources are displayed that are currently assigned to the “released” status.

Memorize

If this checkbox is checked the entered search term is saved, when the dialog is closed. In case nothing is entered in the input field, the value “empty” is saved.

Table

The table of the “manage DNC elements” dialog is already filled while the dialog is being opened, provided that selection is based on programs or orders. In case the “by DNC family” option is selected, the list will only be loaded already while opening the dialog if at least one filter field is filled out.

The “manage DNC elements” dialog shows all available DNC resources:

Maintain setup parameters (SP)

Operation 0100 InjectionMolding

Prod.Order	S6100305	Article	SAR-FMO011-K-5013
------------	----------	---------	-------------------

Workplace 60610 ENG21

Selection	by SD familiy	SP Family	IM_SET
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☐ only status 'released' Search term: ☐ Memorize

Filter	Setup parameter	ESD Familie	File size	Resource status
	ESD-SA-4026-127A	IM_SET	2267	RELEASED
	ESD-SA-4026-127B	IM_SET	2267	RELEASED
	<New>	IM_SET		

Filter: [Article]>SAR-FMO011-K-5013 [Tool]>SG-4026-127 [Machine]>60610 [(Humidity)]

SP record:

Staff Number:

mpciv 07/12/12 23:54:12

Filter fields

The four filter fields are shown here if the selection is based on the “by DNC family” option . Depending on how the DNC family is configured, the fields will be assigned to default values of order and machine. Provided that the “memorize” checkbox is checked, the entered filter criteria will already be assigned the next time the dialog is opened. If, however, a default value is configured for a field in the DNC family, this assignment takes priority.

The filter fields may be entered unless the “read-only” option has been configured. All filter fields can be entered if no OP is logged on. In this case, the input is not mandatory. A configuration in the hytnrcfg.ini dialog can prevent the fields from being released.

[Tnr Konfiguration 0]

DNCFreeSearchReadOnly=On

If the DNC dialog is started from the order sequencing list, filter fields react as it is the case if operations are logged on, as the selected order is transferred as reference.

DNC element

When an entry is selected in the list, the respective DNC element is taken over to the input field. As an alternative, the DNC element may also directly be written into the input field.

The functions that can be started using the buttons down right generally refer to the DNC element displayed in this input field.

Badge number

The staff badge number has been designed for checking authorizations for the functions that are displayed down right in the toolbar and that are described in the following:

-  - Update/refresh
-  - Show DNC resource
-  - Show assigned documents: DNC documents
-  - DNC upload machine → AIP

-  - DNC download AIP → Machine
-  - Lock DNC resource/change status
-  - Release locked DNC resource

2.2 Update

This function updates  the list. The value entered in the “search term” input field is taken into account in this context. The file size of DNC resources available at the AIP is entered in the corresponding column. Furthermore, the status of the DNC resource is displayed.

When loading the list by touching/clicking the corresponding button, it is checked whether or not filter fields configured as “read-only” for the upload are available (configuration of user fields of the DNC family). These fields have to be filled out in order for the list to be loaded.

2.3 Show DNC resource

The file content of the DNC resource is displayed. This function is only available, provided that the DNC-VIS license has been purchased.

2.4 DNC documents

The “DNC documents” function is only available if the “DNC-AEB” license is active (HYDRA-DNC 7.2).

The terminal loads the list of documents assigned to the resource and displays them.

By selecting an entry in the list and clicking/touching the “open document” button, the file is downloaded to the terminal and displayed in an internal or external viewer – depending on the respective file extension.

2.5 Upload

The “upload” function reads a resource from the machine and copies it into the configured target directory. For this purpose, the element has to be released for the upload.

An upload may always be performed if a DNC family is filtered and a <new element> is selected afterwards. At first a dialog appears where element and file name may be entered. Moreover, other filter criteria may be entered. The four lower fields of the upload dialog may be used for this purpose. Configuration of these fields is read out from the DNC family of the resource selected in the DNC dialog.

The fields may be configured as user fields at the client.

A field is only visible if at least one of the options “filter”, “mandatory field at upload”, “pre-assignment at upload” or “read-only” is set.

The field is initialized with default values if the “pre-assignment at upload” option is configured.

The field is grayed out, provided that the “not alterable/read-only” button is set. But the field remains unchanged if the “mandatory field at upload” option is set at the same time.

A new line is displayed in the list after the upload.

The upload is documented as event in the resource history. Two events are entered with the upload. They are to be distinguished by the dialog IDs:

- N: Beginning of the upload; the resource record is entered or changed.
- F: The file pertaining to the record has arrived the target directory.

The two events normally occur in very short intervals. In case the “F” event is missing, the transmission of the file has been interrupted unexpectedly. A download only transfers the file. The record is not changed. Consequently, only one event is entered in the history for the download.

2.6 Download

The “download” function collects a resource from the configured directory and copies it to the machine. Before starting the download, users have to identify themselves by their badge number. It is checked whether or not they are authorized to download programs.

The download process is documented as event in the resource history.

2.7 Lock DNC resource or change status

The resource status may be changed in this dialog.

The status change may be documented as event in the resource history.

Change status of DNC element ESD-SG-8014-127

SP record ESD-SG-8014-127

SP Family IM_SET

Actual status 999

New status 11

Filter

Status	Status designation
11	K-MAINTENANCE
12	OUTSOURCE-K2
13	SCRAPPED
99	BLOCKED

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Comment

Staff Number

☐ Memorize

X Cancel

✓ Change status

07/13/12 00:00:42

2.8 Release DNC resource

When a resource is released its status switches to the status assigned to the processing flag "F". Consequently, the status does not have to be selected or entered.

The status change is documented as event in the resource history.

Release of locked resources:

Release SP element ESD-SG-8014-127A

SP record	ESD-SG-8014-127
SP Family	IM_SET

Actual status	999
----------------------	-----

Comment

Staff Number

9999

☐ **Memorize**

Cancel

Release SP record

07/12/12 23:59:25

3 Local Configuration File ctaip.ini

The most important hardware and system settings are defined for each terminal in the CTAIP.INI file of the c:\ctaip directory.



Changes to the configuration file ctaip.ini are only enabled after rebooting the terminal software.

3.1 Basic configuration

Entry	Comment
Section [system]	
Usr=21	Distinct terminal number
Hypath=d:\hydra\ Hypath=/usr/hydra/	HYDRA server path: Windows NT: -> DOS notation, the drive is the local drive of the server on which HYDRA or xMES is installed Unix: -> Unix notation
Hostname=192.9.200.24	Internet address of the server
Offlinetimeout=600	In offline mode, the interval after which online access should be attempted the next time. The interval is specified in seconds
Showcursor=on	Show or hide mouse pointer in terminal application: on: mouse pointer active off: mouse pointer inactive
Loadfile=ctnet\win\ctaip.txt	Configuration file for downloading the application from the server. The path is relative to the server directory (i.e. within "hypath")
Watchdog=on	ON: Watchdog is activated OFF: Watchdog is not activated
Demo=off	'on': Offline demo mode; always off in the production environment
parameters=-t	The -t parameter switches off the virtual keyboard
TMOUT_C=xxx	Timeout for CONNECT to the server If not specified, default = 10 seconds → Increase to 20 seconds for routing
TMOUT_S=xxx	Timeout for SEND to the server If not specified, default = 10 seconds → Increase to 20 seconds for routing
TMOUT_R=xxx	Timeout for RECEIVE of the server If not specified, default = 120 seconds

Entry	Comment
TMOUT_F=xxx	Timeout for FILESERVER operations to the server If not specified, default = 10 seconds → Increase to 20 seconds for routing
Section [barcode]	Configuration of customized barcode prefixes.
BarKenn90=MNR4 ... BarKenn99=ANR3	BarKenn90 > defines the prefix (here: 90); The ID from the dialog (= acronym) is assigned.
Section [comports]	
Com1=0 com2=MSS com3=BAR Com3=LEGIC Com4=RFLESER	Assignment of serial interfaces to the connected devices MSS – machine interface BAR, LEGIC, RFLESER – various reading devices
Section [MSS-INIT]	
ZAEHLER= 1 2 3 4 5 6 7 8	Assignment of physical inputs of the MSS (machine interface) to logical counters (ZAEHLER) according to configuration: The first connector (labeled “0” on the MSS) corresponds to the logical counter no. 1 Please note: MSS1 has only 8 inputs If digital inputs are also to be used with MSS1, configuration should be changed as follows: ZAEHLER= 1 2 3 4 IN= 5 6 7 8
IN= 9 10 11 12 13 14 15	Assignment of physical MSS inputs to logical inputs as per configuration: The ninth connector (labeled “8” on the MSS) corresponds to the logical input no. 1
CHARGE= 5 6	For batch recording: Inputs for automatic batch changes In this case, the connectors 5 and 6 are the inputs for the automatic batch change.
MSSZyklusBerechnung=ON MSSZyklusReferenz=0	Activates reading out of cycle time values from the machine interface (MSS). Reference for cycle calculation (smallest time unit of the MSS). The default value is 0, if the parameter is not specified. 0 corresponds to 100 ms. 2 corresponds to 20 ms.
CalculateCycle=ON	The terminal itself calculates the actual cycle. If the connected control does not provide a determined actual cycle, then a calculated actual cycle can be displayed: This calculated value is also available for DS100 terminals. Please note: The calculation cannot provide exact values.
WochenEnde_ProdCheck=ON	This function prevents the weekend automatic from affecting the “production” status and the workplace from being set to status 999. ON is set by default In case WochenEnde_ProdCheck=OFF, the automatism switches to status 999.

Entry	Comment
sFrom999ToNotAttributed=OFF	Only affects HYDRA-MDE machines with operation mode = "no monitoring". If the weekend automatic function is enabled this option prevents the machine from switching to the "not assigned" status when the weekend automatic ends. Reasons: The "not assigned" status may not be set manually for machines that are configured with the "no monitoring" option. The "not assigned" status is normally only set for machines with operation mode = "cyclic monitoring" or "monitoring by operating signal".
Section [ext. software]	
Button=Editor WindowName=Editor SearchParts=On	Configuration of the button in the top line: A previously started program can be called to the foreground at the push of a button. Button: button caption WindowName: Name of the program (e.g. from the taskbar). SearchParts=On: parts of WindowName are sufficient SearchParts=Off: WindowName must be entered completely. The option "SearchParts=On" is recommended for programs such as MSWord that change the title bar subject to the document that is currently being loaded.
ProgFileName=c:\Programme\wincmd\Wincmd32.exe	The program that is started if the program mentioned above cannot be called to the foreground.
AutoStart=on	This option starts the program (ProgFileName) when starting the terminal program.
Section [PDV]	
Modus=PDV Modus=PDV,BDE	Operation as interface IOP↔DOS terminal (standard) Operation as shop floor terminal with PDV
Terminals=121,122,123	CTDOS terminals "connected" by LAN. Only required with Mode=PDV
PDVTerminalDir=c:\hsrv\spool\ PDVIOPTDir=c:\IOPSim\	Directory for communication with CTDOS terminals Directory for communication with the IOP
For debug purposes only: InfoFenster=100 SlowDown=600	Number of lines of the "current" window (presentation of last PDV actions) Slow down, to make events "visible"
;Supported barcodes	
FieldWNRBarcodeOnly=Y	If this entry is set the tool number may only be entered using a scanner.
FieldNestBarcodeOnly=Y	If this entry is set the cavity number may only be entered using a scanner.
FieldNummBarcodeOnly=Y	If this entry is set the number may only be entered using a scanner.
FieldKNRBarcodeOnly=Y	If this entry is set the badge number may only be entered using a scanner.
BarcodeWNR=	This field specifies which acronym is entered into the tool number field by the scanner.

Entry	Comment
BarcodeNest=	This field specifies which acronym is entered into the cavity number field by the scanner.
BarcodeNumm=	This field specifies which acronym is entered into the number field by the scanner.

4 Central Configuration File hytnrcfg.ini

This file includes different configurations for all or single terminals at a central place.

Each section is available in a generally accepted version

[section 0].

However, entries included in this section can be overwritten by entries in a terminal-specific section [section <TNR-USER>]

<TNR-USER> = HydraUser = Terminal number + 2000 e.g. 2010,2101,..) for exactly one terminal/HYDRA User



The hytnrcfg.ini file is loaded from the server every time the terminal is started.

Section / Entry	Comment
	→→→
[Tnr configuration 0]	
FollowExternStatus=on	Transfer of machine statuses when reloading machine list Useful if status change is set by PDM or another terminal
[Terminal->Installation 0]	
InstallFonts=on	If this is set to "off" fonts will not be installed during the restart. ON=DEFAULT
OnlyInstallFontsAfterDownload=false	"InstallFonts=on": If true then fonts will only be installed directly after a download. If false then fonts will be installed every time the terminal is restarted. (false = DEFAULT)
InstallTvicport=on	If "off" the LPT driver "tvicport.sys" will not be installed. It is required for HYDRA-ZKS. ON = DEFAULT
[Terminal->USR 0]	

AttachedApplication=First	This configuration checks whether or not an application is connected in Windows that matches the file extension of the document to be displayed from the OP info dialog. If there is such an application, it will be used for displaying the document. If there is no connection, viewers configured in ctaip.ini (→ [ext. software]) and internal viewers will be used. In case, an extension is completely unknown it is attempted to display it as text Different settings may be configured: First → search for connected application first AfterUserViewer → If a UserViewer is configured this one overrides the connected application (also applies for ExcelViewer, WordViewer and PowerpointViewer) Last → Only if no ctaip.ini assignment is found for the file extension, then the connected assignment will be searched for (default). Off → Connected application is never searched.
HTTPBrowser=standard	Viewing of documents (via OP info): If documents are configured with a path of the <u>type</u> "http", the file will not be downloaded to the terminal, but the link will only be transferred to a browser. The default browser for the terminal is htmview3.exe, as this one can be operated by touchscreen. If this entry is set, the default browser configured in Windows will be used.
SupressErrorMessage=70012	Suppress message "material is not planned"
[SignatureRecording->User 0]	
ManualBadgeInput=true	This configuration specifies whether or not the field "user" can be edited in the terminal (by default: no editing) true → activates keyboard input for the "user" field in the terminal

Transparency=255	The signature dialog can also be transparent. 255 → Signature dialog is 0 % transparent (not transparent) 1 → Signature dialog is 99% transparent (maximum transparency) (Default = 155) Available as of CTAIP (V# 2.0.2.25) / CTWIN (V# 7.2.5.99)
ShowPosition=TR	The position of the signature dialog can be adjusted as follows: TL Top – Left TM Top – Middle TR Top – Right ML Middle – Left MM Middle – Middle (Default) MR Middle – Right BL Bottom – Left BM Bottom – Middle BR Bottom – Right Available as of CTAIP (V# 2.0.2.25)
USE_SERVICE_ACCOUNT=1	0 (default) SSO: do not use ServiceAccount (requires the terminal to be started with the "user" domain (SSO)). Please note: ServiceAccount=1 can only be used if all users are in the "root" domain. SubDomain users are not supported.
SIGNATURE_1_USER_TYPE=REPORTING_USER_READONLY	REPORTING_USER_READONLY The tab identifying users via the Windows user is activated and assigned to "user" by default. The "user" field is read-only. This requires, however, that in the HYDRA HR master the "SSO" option is set for all users logging in. Otherwise, successful authentication is impossible. REPORTING_USER_CHANGEABLE The tab identifying users via the Windows user is activated and assigned to "user" by default. The "user" field can be modified. This requires, however, that in the HYDRA HR master the "SSO" option is set for all users logging in. Otherwise, successful authentication is impossible.

SIGNATURE_1_LOGON_TYPE=HYDRA	"" / Not set / "EMPTY" There is also an alternative login procedure. HYDRA The tab identifying users via the Windows user is blocked. The HYDRA user must be used for identification purposes. This requires, however, that in the HYDRA HR master all users logging in are created and that the "SSO" option is not set. Otherwise, successful authentication is impossible. ACTIVEDIRECTORY The tab identifying users via the HYDRA user is blocked. The Windows user must be used for identification purposes. This requires, however, that in the HYDRA HR master the "SSO" option is set for all users logging in. Otherwise, successful authentication is impossible. MIXED_BUT_UNIQUE Either the HYDRA or Windows login procedure is available, subject to whether or not the "SSO" option is set for the registered user in the HYDRA HR master. "SSO" enabled → Windows only "SSO" disabled → HYDRA only
SIGNATURE_2_LOGON_TYPE=HYDRA	Identical to SIGNATURE_1_LOGON_TYPE (see above)
ExtendedSignatureRecording=true	Used for signatures with the terminal in the area of quality data collection.

4.1 Layout configuration

Entry		Comment
Section and/or	[terminal configuration 0] [terminal configuration 2XXX];	(general configuration) (2XXX terminal-specific configuration)
AUTO-CONFIRM-UHR-ERROR-MESSAGE=TRUE		In case of an error in reading the clock (e.g. after coming out of standby mode), this configuration makes sure that the time is accepted without having to confirm a dialog. Afterwards the terminal time will be synchronized with the server time using a PDM command.
SUPPRESS-MAXIMUM-NUMBER-OF-MACHINES-WARNING=ON		As of ctaip V# 2.0.2.23 Prevents the warning after restarting the terminal if more than 32 machines are assigned to the terminal (static/dynamic). (Default = OFF)

Entry	Comment
NetRuntimeMode=2	As of ctaip V# 2.0.2.50: Alternative calculation of the target quantity since logon: The net run time is not calculated from the times when the production lock is enabled (PSperre=green) but only from the shift times less the shift breaks. Consequently, it can also be displayed, even if the terminal program has been restarted.
Section [QRD-PRINTER->TICKET 0] ;(general configuration) [QRD-PRINTER->TICKET 2xxx] ;(2XXX configuration for a specific terminal)	
COMPLETE-ABSENCE-OF-LOCAL-MNR- DATA-FOR-EVENT=< Events >	Reloads the machine row for the configured <Events>, if it is not available locally => This configuration might be required/necessary for a group workplace without machine assignment.
COMPLETE-ABSENCE-OF-LOCAL-ANR- DATA-FOR-EVENT=< Events >	Reloads the order row for the configured <Events>, if it is not available locally ➔ This option has been implemented to access order data within the master data, e.g. when logging an order on.
COMPLETE-...-EVENT=< Events > COMPLETE-...-EVENT=#ALL# COMPLETE-...-EVENT=A_AN A_P_AN	Explanation on the configuration of <Events> ➔ Using <#ALL#> the row (ANR/MNR) that is not available is reloaded for any event. ➔ <A_AN A_P_AN> restricts reloading of information to the specified events. The ID <DLGFAM> is preferred to the ID <DLG> in order to identify the <Event>.